

ABSTRACT

This report presents the design and implementation of a desktop-based Smart Parking Management System developed using Java Swing, JDBC, and MySQL. The primary objective of the system is to simplify and organize vehicle parking management by providing an efficient and user-friendly interface, thereby reducing manual effort and improving the maintenance of parking records.

The application supports essential operations such as adding, viewing, updating, and deleting parking records through a graphical user interface. Each parking record contains details such as vehicle number, owner name, vehicle type, and parking slot information. The system is structured using a layered architecture consisting of Data Transfer Object (DTO), Data Access Object (DAO), Service, and User Interface (UI) layers. This separation of concerns enhances modularity, maintainability, and scalability of the application.

The DAO layer is responsible for handling database operations using JDBC, enabling secure and efficient interaction with the MySQL database. The service layer incorporates business logic and input validation to ensure the accuracy and consistency of parking data. The UI layer, developed using Java Swing, provides an intuitive platform for users to interact with the system and perform operations seamlessly.

This project demonstrates the integration of a Java-based graphical interface with a relational database system and highlights the importance of structured software design in developing reliable and maintainable parking management applications.